

Understanding Randomization in Experiments

Aodi Cheng

November 18, 2024

Solving Confounding: Experiments

► Confounding Problem:

$$Y = f(X, W) + \epsilon$$

- X : Treatment (e.g., knowledge of Biden/Trump being vaccinated).
- W : Confounder (e.g., trust in leadership).
- ϵ : Random error.

► Goal: Estimate causal effect of X on Y :

$$\frac{\partial Y}{\partial X}$$

► Without randomization: W influences both X and Y :

$$\text{Cov}(W, X) \neq 0 \quad \Rightarrow \quad \text{biased estimate of } \frac{\partial Y}{\partial X}.$$

Randomization and its Effects

- ▶ **Randomization ensures independence:**

$$\text{Cov}(W, X) = 0$$

- ▶ **Result:**

$$Y = \beta_0 + \beta_1 X + \epsilon \quad \text{where } \beta_1 = \frac{\partial Y}{\partial X}.$$

- ▶ **Key Mechanism:**

- ▶ Treatment X assigned randomly.
- ▶ No systematic relationship between W (confounder) and X .

Internal and External Validity

- ▶ **Internal Validity:**

- ▶ Low confounding.
- ▶ Randomization ensures causal interpretation of β_1 .

- ▶ **External Validity:**

- ▶ Applicability of findings to the real world.
- ▶ Example: Does Biden's vaccination influence the general population when communicated via TV?

Causal Question: Vaccine Hesitancy Study

Causal Question: Does knowledge of a president being vaccinated encourage individuals to vaccinate?

- ▶ **Independent Variable (X):** Awareness of president's vaccination.
- ▶ **Dependent Variable (Y):** Willingness to vaccinate.
- ▶ **Confounding Problem:**

$$\text{Cov}(W, X) \neq 0$$

- ▶ **Randomization ensures independence:**

$$\text{Cov}(W, X) = 0$$

Small Groups Activity: Vaccine Hesitancy Study

1. Read the article: *Can vaccine hesitancy be reduced by a president's encouragement?*
2. **Discussion Questions:**
 - ▶ What makes this an experiment?
 - ▶ What is the “treatment”? Is *X* the *only* thing changing between treatment and control groups?
 - ▶ How does the experiment address confounding (internal validity)?
 - ▶ What concerns do you still have about this experiment?

Group Activity: Proposing a New Experiment

Imagine you have unlimited resources (and no ethical constraints):

1. What units (e.g., people, universities) would be randomized?
2. What would the treatment be? What would be different in the control group? How would you ensure only X is changing?
3. How might this experiment lack external validity?

Group Sharing:

- ▶ Share your proposed experiment.
- ▶ Focus on the intervention (how you only change X).
- ▶ Class discussion: Raise questions about external validity.